



Problem solving with similarity and Pythagoras' theorem

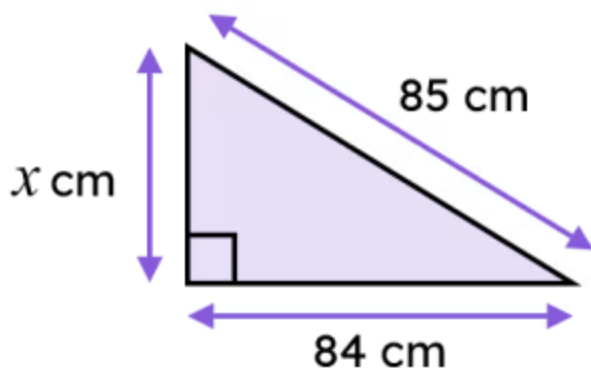
- 1 Alex rearranges a correct Pythagoras' theorem equation for a triangle. Which of these statements are correct? Tick **2** correct answers

$$65^2 - 25^2 = r^2$$

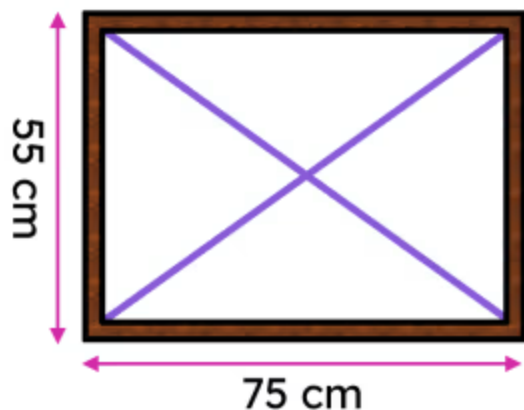
side P side Q side R

- ☒ The hypotenuse of the triangle is side P
- ☐ The hypotenuse of the triangle is side Q
- ☐ The hypotenuse of the triangle is side R
- ☒ Side R has length of 60 units.
- ☐ Side R has length of 69.64 units.

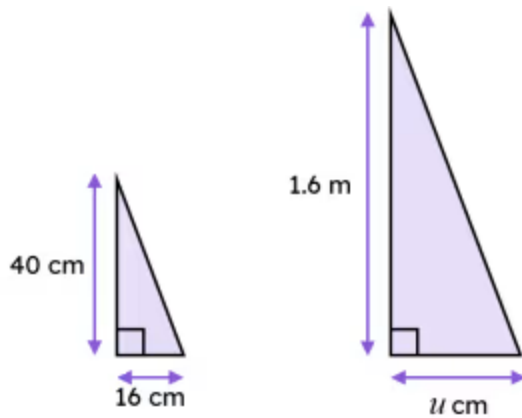
- 2 The perimeter of this right-angled triangle is 182 cm. Fill in the blank



- 3 Adhesive tape is placed along the two diagonals of this rectangular picture frame. How much adhesive tape is needed, rounded to the nearest cm? 186 cm of tape. Fill in the blank



- 4 Andeep and Sofia need to split a 3.6 metre roll of adhesive tape in the ratio 7 : 5. The length of the adhesive tape that Andeep receives is 210 centimetres. Fill in the blank
- 5 These two triangles are similar to each other. The side marked u cm is 64 cm long. Fill in the blank



- 6 An isosceles triangle has an angle of 22° . Which of these are possible sizes of one of the other angles? Tick **3** correct answers

☐ 11°

☒ 22°

☐ 45°

☒ 79°

☒ 136°

